

Reliability Analysis of a Stochastic Network via Weyl-Sequence Simulation

Background: The Middle-Square Weyl Sequence (MSWS) generator keeps *two* running 32-bit states: a value x (the “middle-square” state) and a value w (the “Weyl” state), both taken modulo 2^{32} . Starting from $x_0 = 0$ and $w_0 = 0$, each new variate is produced by the following update, using the fixed increment $c = 0xb5ad4ec5$:

One generator step (produces the next state x_{n+1} from x_n, w_n):

1. Advance the Weyl state: $w_{n+1} = (w_n + c) \bmod 2^{32}$.
2. Square-and-add: $x_{n+1} = (x_n^2 + w_{n+1}) \bmod 2^{32}$.
3. Extract the output bits by discarding the low 16 bits of x_{n+1} :

$$U_n = \frac{\lfloor x_{n+1}/2^{16} \rfloor}{2^{16}} \in [0, 1).$$

(This is equivalent to the bit-shift $x_{n+1} \gg 16$ described by right-shifting; you may implement it with plain integer division, no bitwise operators required.)

Task 1

[3]

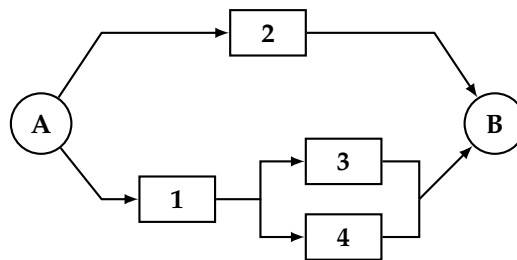
Implement the Middle-Square Weyl Sequence generator described above, with initial seed $x_0 = 0$ and initial Weyl state $w_0 = 0$. It should return a list of N variates $U_0, \dots, U_{N-1}$.

```
def msws_generator(n_samples, x0=0, w0=0, c=0xb5ad4ec5, m=2**32):
    # your implementation
```

Task 2

[5]

Consider a 4-component stochastic network trying to send a signal from node A to node B.



Each component $i \in \{1, 2, 3, 4\}$ fails independently after time $X_i \sim \text{Exponential}(\text{mean} = \beta_i)$ in days:

$$\beta_1 = 10, \quad \beta_2 = 8, \quad \beta_3 = 7, \quad \beta_4 = 5$$

Build a Monte Carlo simulation of $N = 10,000$ replications to estimate the probability p that the overall system survival time $Y \geq 7$ days, using a random number assignment scheme from the Middle-Square Weyl Sequence generator.

Inverse-CDF Formula: To convert a uniform random variate $U \sim \text{Uniform}(0, 1)$ to an exponentially distributed lifetime $X \sim \text{Exponential}(\text{mean} = \beta)$, use:

$$X = F^{-1}(U) = -\beta \ln(1 - U)$$

Steps:

1. **Generate Stream:** Produce a $U(n)$ stream, long enough to satisfy the scheme in the next step.
2. **Stream Assignment:** Select uniform variates $U(n)$ for the four component lifetimes of each replication $k \in \{0, \dots, 9999\}$ under the following scheme:

- **Scheme:** For replication k , set

$$U_1 = U(4k), \quad U_2 = U(4k + 1), \quad U_3 = U(4k + 2), \quad U_4 = U(4k + 3).$$

3. **Variate Transformation:** Convert the assigned uniform variates U_1, U_2, U_3, U_4 into component lifetimes X_1, X_2, X_3, X_4 using the inverse-CDF formula, with $X_i \sim \text{Exponential}(\text{mean} = \beta_i)$.
4. **Simulate Network:** For any two subsystem lifetimes T_a and T_b , the survival time of their *series* and *parallel* combinations are given by:

$$T_{\text{series}}(T_a, T_b) = \min(T_a, T_b), \quad T_{\text{parallel}}(T_a, T_b) = \max(T_a, T_b).$$

Using only these two combination rules, work out and implement the correct sequence of operations that computes the overall system survival time Y_k for the topology shown above, applied to the four component lifetimes X_1, X_2, X_3, X_4 of replication k .

5. **Calculate Reliability:** Compute the estimated probability of survival past 7 days,

$$\hat{p} = \frac{1}{N} \sum_{k=0}^{N-1} \mathbb{I}(Y_k \geq \theta).$$

Task 3**[2]**

The company maintains this network but can afford to buy exactly **one spare component**, identical in reliability to one of components 1, 2, 3, or 4, and install it as a redundant unit in **parallel** with that *one chosen component* (leaving the rest of the topology unchanged). Based on the structure of the network in Task 2, not on any new simulation or formula derivation, decide which single component (1, 2, 3, or 4) the spare should be added to in order to improve the 7 day survival probability p the most, and justify your choice.